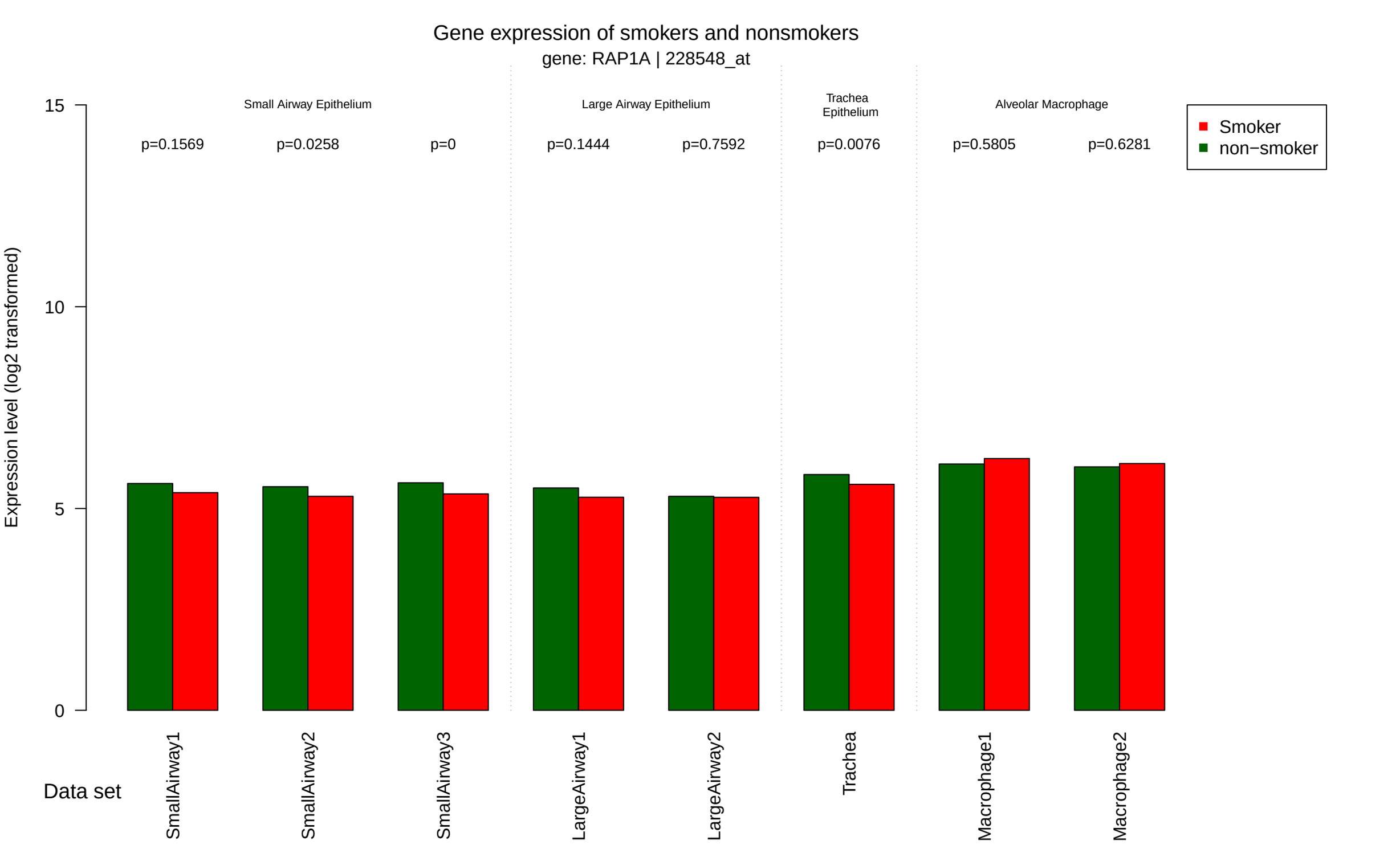
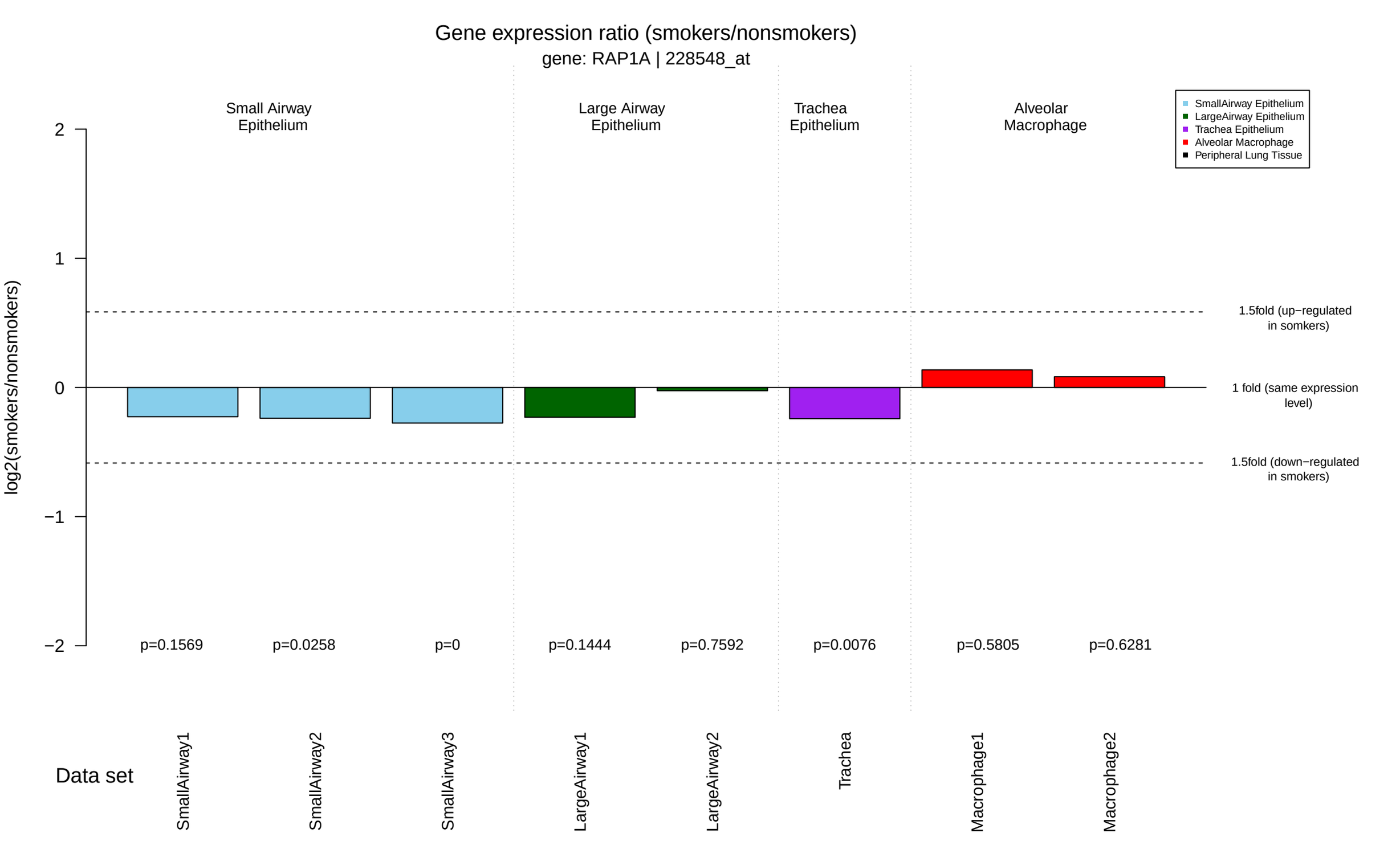
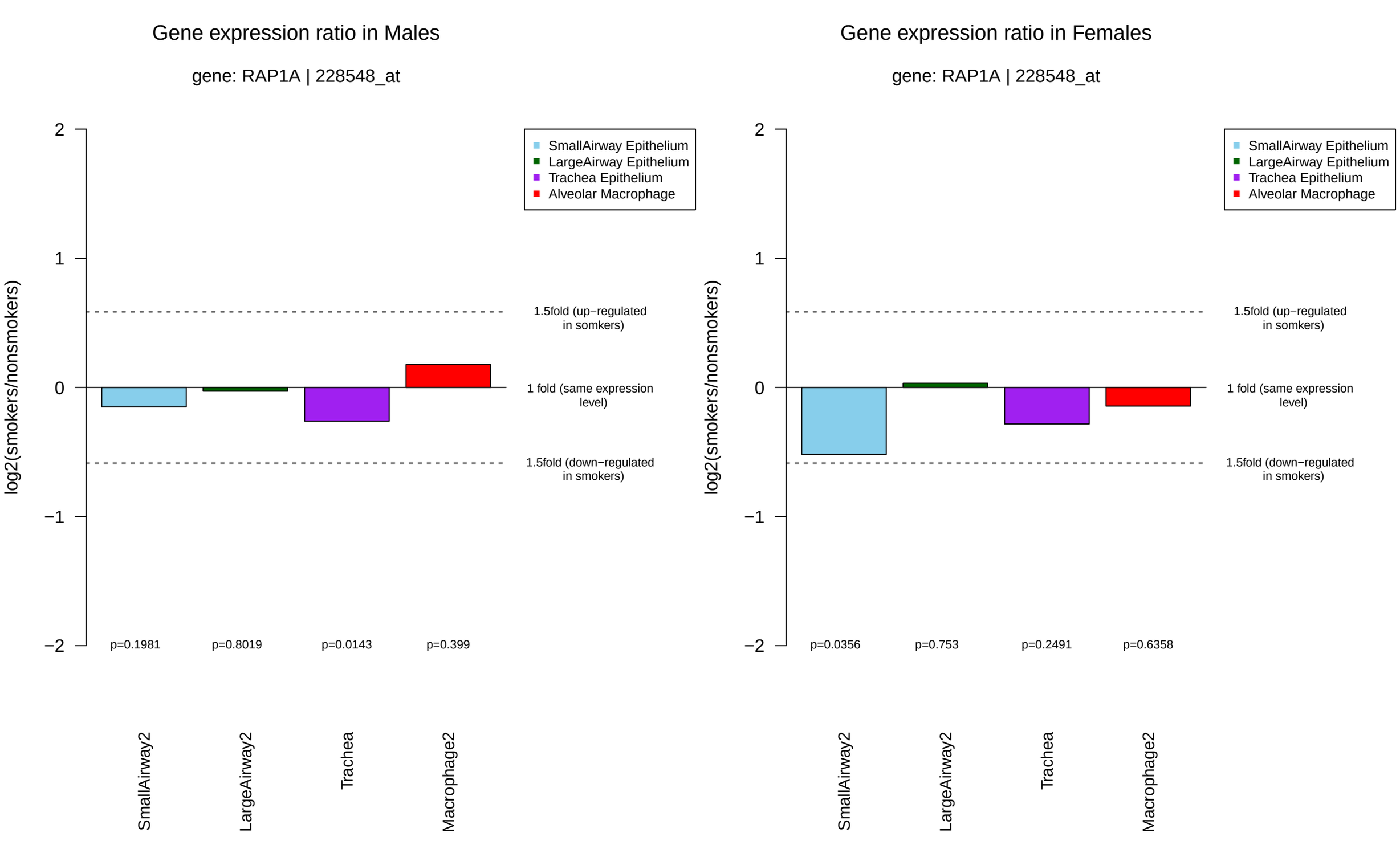
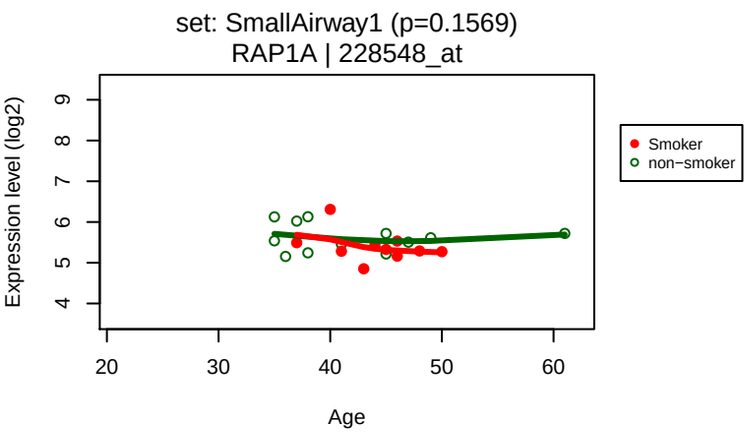
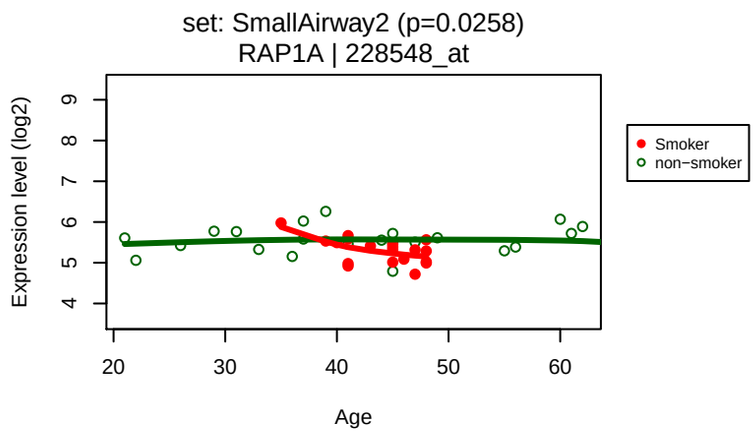
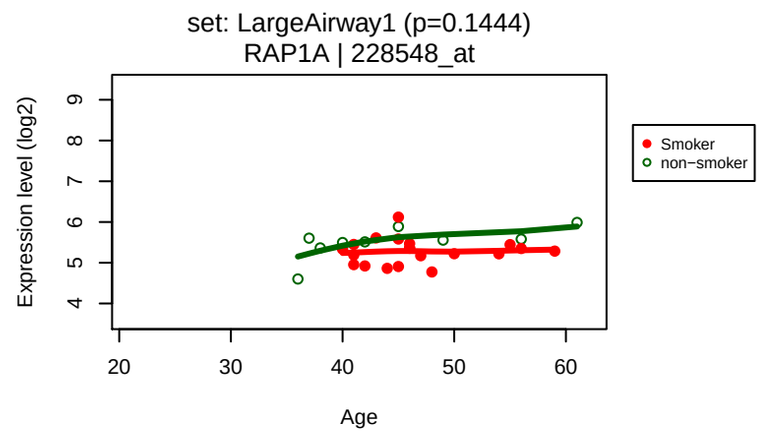

Gene expression of smokers and nonsmokers
gene: RAP1A | 228548_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1569
p=0.0258
p=0
p=0.1444
p=0.7592
p=0.0076
p=0.5805
p=0.6281
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

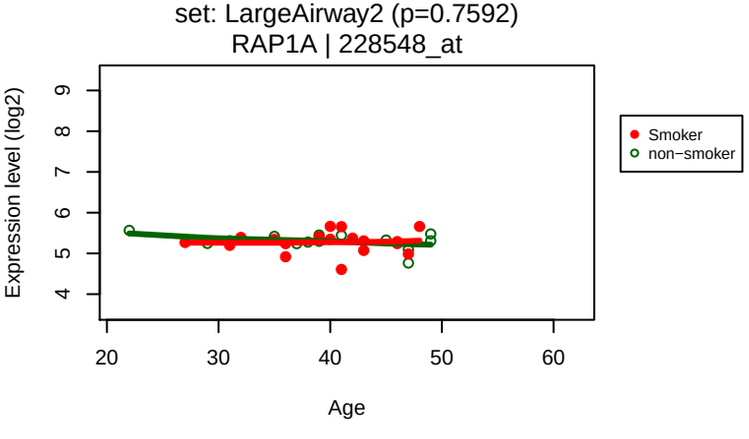

Gene expression ratio (smokers/nonsmokers)
gene: RAP1A | 228548_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1569
p=0.0258
p=0
p=0.1444
p=0.7592
p=0.0076
p=0.5805
p=0.6281
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

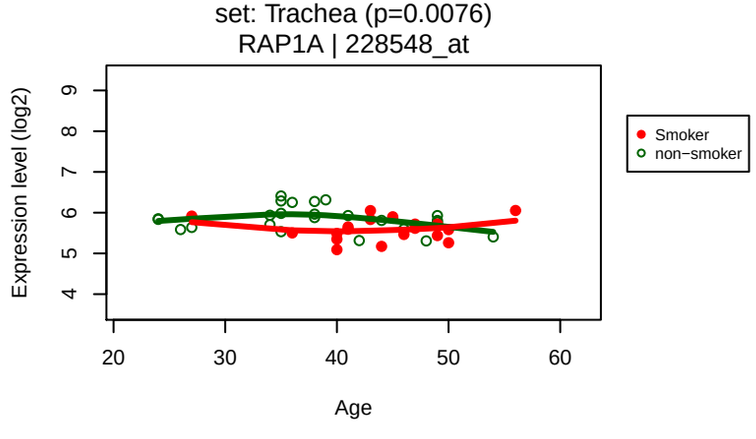

Gene expression ratio in Males
gene: RAP1A | 228548_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1981
p=0.8019
p=0.0143
p=0.399
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAP1A | 228548_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0356
p=0.753
p=0.2491
p=0.6358
SmallAirway2
LargeAirway2
Trachea
Macrophage2

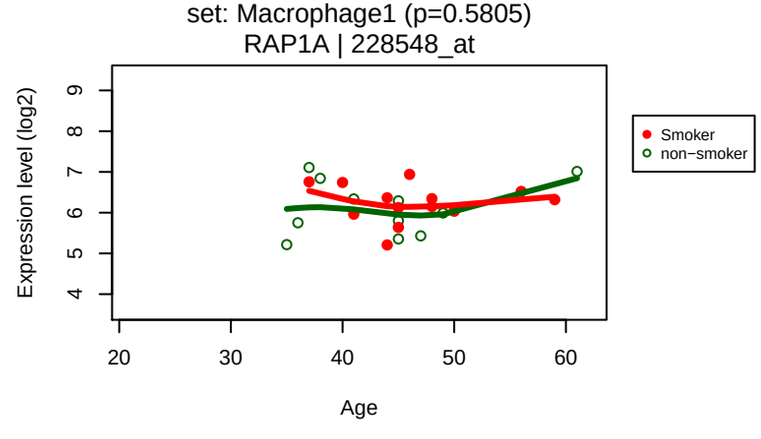

set: SmallAirway1 (p=0.1569)
RAP1A | 228548_at
Expression level (log2)
Age
Smoker
non−smoker

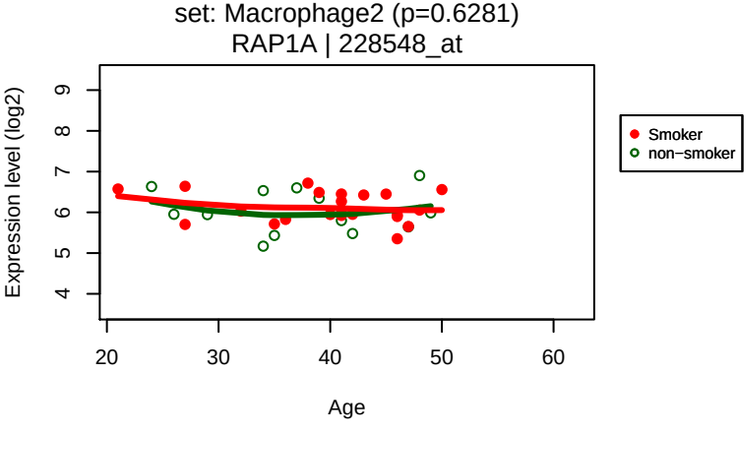

set: SmallAirway2 (p=0.0258)
RAP1A | 228548_at
Expression level (log2)
Age
Smoker
non−smoker

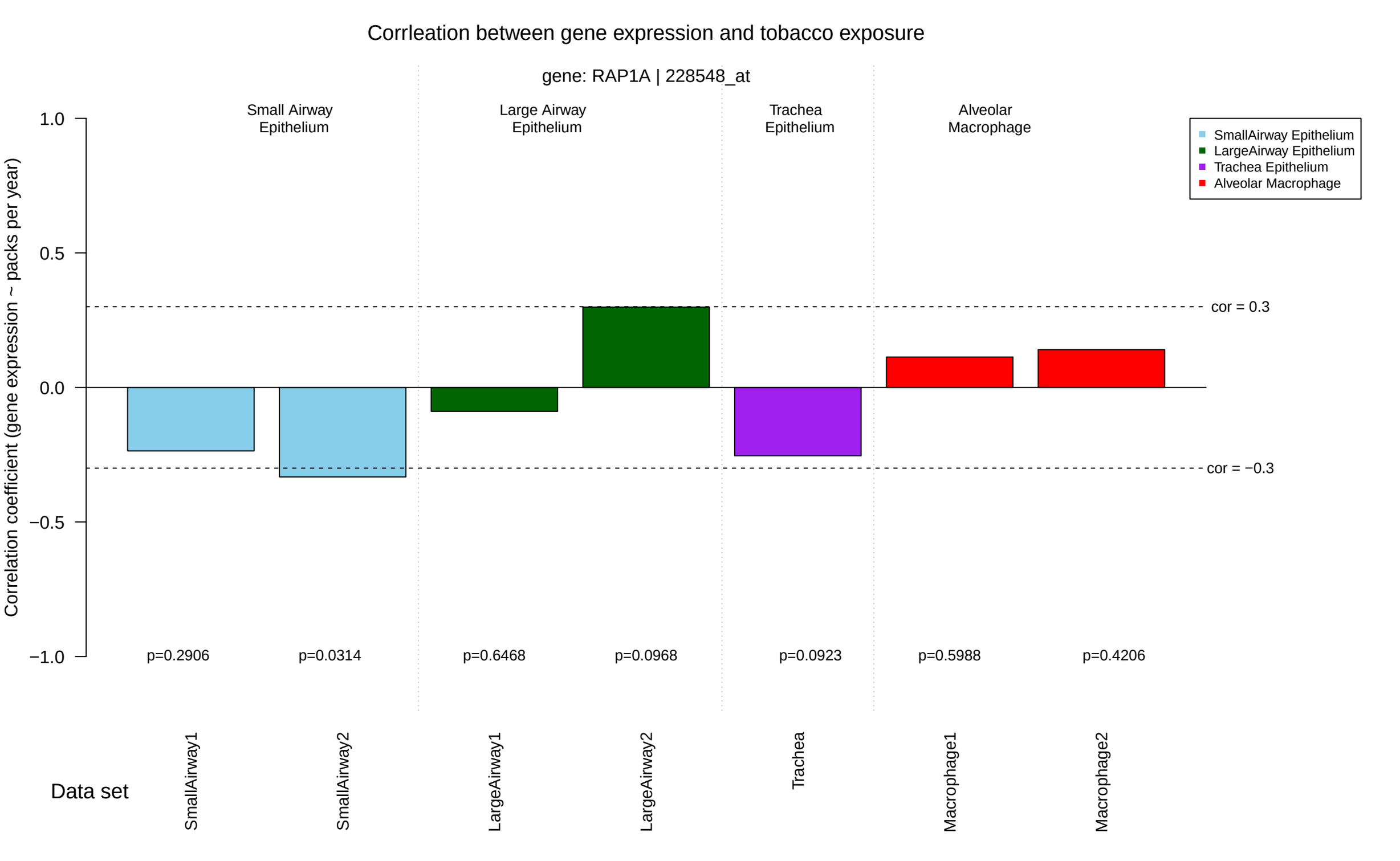

set: LargeAirway1 (p=0.1444)
RAP1A | 228548_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7592)
RAP1A | 228548_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0076)
RAP1A | 228548_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5805)
RAP1A | 228548_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6281)
RAP1A | 228548_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAP1A | 228548_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2906
p=0.0314
p=0.6468
p=0.0968
p=0.0923
p=0.5988
p=0.4206
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2